PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T42 Classified

Cross-session • chapter-organised candidate

16 classified items • 60 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Unit 2 • Chapter OL-T42 • 16 items • 60 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Circle Theorems

Unit 2 • 60 marks • 16 item(s)

June 2024 | 4MA1-2H Q21 | 5 marks

Use radius perpendicular to tangent

November 2023 | 4MA1-1H Q14 | 3 marks

Complete a multi-theorem circle-angle chain

November 2023 | 4MA1-1H Q22 | 6 marks

Use radius perpendicular to tangent

June 2023 | 4MA1-2H Q7 | 3 marks

Use radius perpendicular to tangent

June 2023 | 4MA1-2H Q14 | 4 marks

Use angles in the same segment

November 2024 | 4MA1-2H Q16 | 5 marks

Complete a multi-theorem circle-angle chain

January 2022 | 4MA1-2H Q16 | 4 marks

Complete a multi-theorem circle-angle chain

January 2022 | 4MA1-2H Q19 | 3 marks

Use the internal intersecting-chords product

January 2020 | 4MA1-2H Q13 | 4 marks

Complete a multi-theorem circle-angle chain

January 2021 | 4MA1-2H Q13 | 2 marks

Complete a multi-theorem circle-angle chain

January 2021 | 4MA1-2H Q13 | 1 marks

Complete a multi-theorem circle-angle chain

June 2021 | 4MA1-1H Q15 | 4 marks

Complete a multi-theorem circle-angle chain

November 2021 | 4MA1-2H Q11 | 3 marks

Complete a multi-theorem circle-angle chain

May-Nov 2020 | 1H Q16 | 4 marks

Complete a multi-theorem circle-angle chain

January 2023 | 2H Q14 | 5 marks

Complete a multi-theorem circle-angle chain

... 1 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Question 24
4MA1-2H Q21

ORIGINAL QUESTION
5 marks

21 The diagram shows a square *ABCD* and a circle.

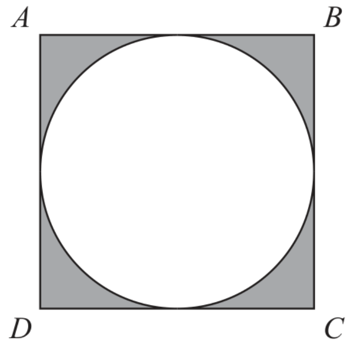

Diagram **NOT** accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is $80\,\text{cm}^2$

Work out the length of *AC*
Give your answer correct to 3 significant figures.

.... cm

Worked Solution 24

Square and inscribed circle

MODULAR UNIT 2
5 marks

Area

$$(2r)^2 - \pi r^2 = 80 \Rightarrow r^2 = 80/(4 - \pi).$$

Diagonal

$$AC = 2r\sqrt{2} = 27.305\ldots$$

Final answer

$$27.3 \text{ cm}$$

Question 11

4MA1-1H Q14 demand

ORIGINAL QUESTION UNIT 2

3 marks

14

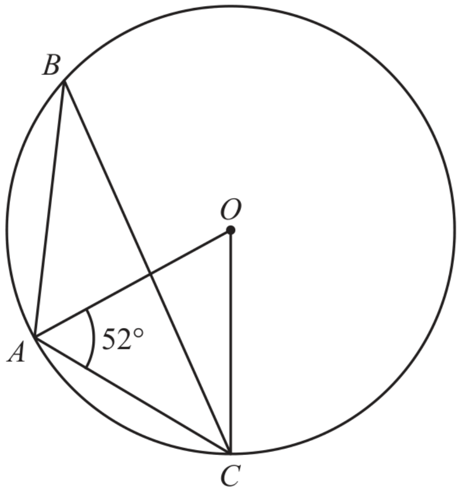

Diagram **NOT** accurately drawn

A, *B* and *C* are points on a circle, centre *O*

Angle $OAC = 52°$

Find the size of angle *ABC*
Give reasons for your working.

....°

Worked solution 11

Angle at the circumference

MODULAR UNIT 2
3 marks

Demand – Angle at the circumference

Find the central angle

$$\angle AOC = 180^\circ - 2(52^\circ) = 76^\circ.$$

Use the circle theorem

$$\angle ABC = \frac{1}{2}\angle AOC = 38^\circ.$$

State reasons

$$OA = OC (radii), so base angles are equal; angle at centre is twice angle at circumference.$$

Final answer

38°

Question 17

4MA1-1H Q22 demand

ORIGINAL QUESTION UNIT 2

6 marks

22 [In this question 1 cm = 1 unit on the x-axis and 1 cm = 1 unit on the y-axis]

P is a point on a circle with centre (0, 0)
The coordinates of P are (8, –10)

The line **L** is the tangent to the circle at the point P
L crosses the x-axis at the point Q and crosses the y-axis at the point R

Work out the length of RQ
Give your answer correct to 3 significant figures.

.... cm

Worked solution 17

Length between tangent intercepts

MODULAR UNIT 2

6 marks

Demand – Length between tangent intercepts

Find radius gradient

$$m_{OP} = (-10 - 0)/(8 - 0) = -5/4.$$

Find tangent gradient

$$m_L = 4/5 \text{ since } m_{OP}m_L = -1.$$

Find intercepts

$$y + 10 = \frac{4}{5}(x - 8) \implies R = (0, -16.4),\ Q = (20.5, 0).$$

Distance

$$RQ = \sqrt{20.5^2 + 16.4^2} = 26.3 \text{ cm (3 s.f.)}.$$

Final answer

26.3 cm

Question 21

4MA1-2H Q7

ORIGINAL QUESTION

3 marks

7

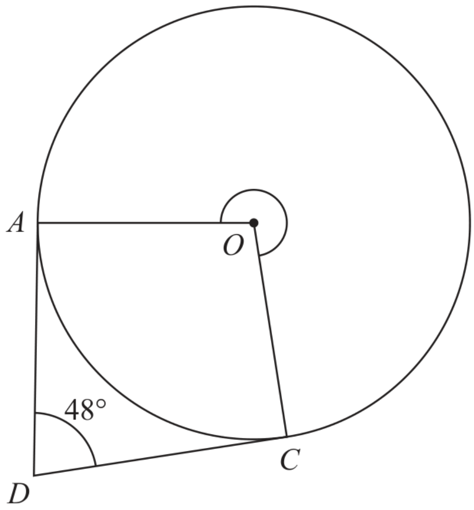

Diagram **NOT** accurately drawn

A and *C* are points on a circle, centre *O*
DA is the tangent to the circle at *A* and *DC* is the tangent to the circle at *C*

Angle $ADC = 48°$

Work out the size of reflex angle *AOC*

.... °

Worked Solution 21

Reflex angle between tangents

MODULAR UNIT 2
3 marks

Spot the right angles

$$OA \perp AD \text{ and } OC \perp CD, \text{ so the two tangent angles are } 90^\circ.$$

Find the reflex angle

$$\angle AOC_{\text{reflex}} = 90 + 48 + 90 = 228^\circ.$$

Final answer

$$228^\circ$$

Question 25

4MA1-2H Q14

ORIGINAL QUESTION
4 marks

14

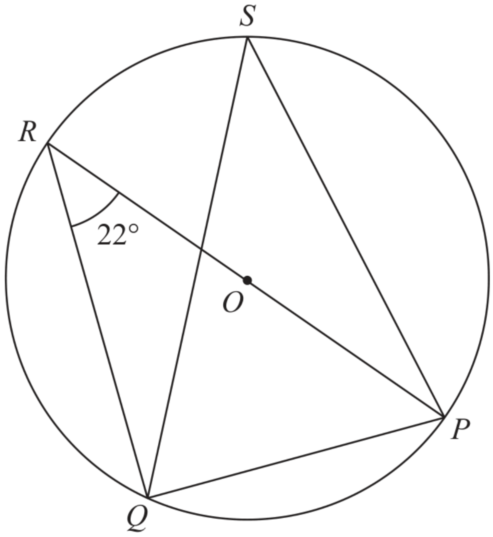

Diagram **NOT** accurately drawn

P, Q, R and S are points on a circle, centre O
ROP is a diameter of the circle.
Angle $PRQ = 22°$

(a) (i) Find the size of angle RQP

....°

(1)

(ii) Give a reason for your answer.

....

....

(1)

(b) (i) Find the size of angle PSQ

....°

(1)

(ii) Give a reason for your answer.

....

....

(1)

Worked Solution 25

Diameter and equal-segment circle angles

MODULAR UNIT 2

4 marks

Diameter angle

$$ROP \text{ is a diameter, so } \angle RQP = 90^\circ \text{ (angle in a semicircle).}$$

Same chord

$$\angle PRQ = 22^\circ \text{ and } \angle PSQ \text{ stand on chord } PQ, \text{ so } \angle PSQ = 22^\circ.$$

Final answer

$$\angle RQP = 90^\circ, \quad \angle PSQ = 22^\circ$$

Original question 28

4MA1-2H Q16 M01, M02, M03, M04

MODULAR UNIT 2
5 marks

16 A, B, C and D are points on a circle, centre O

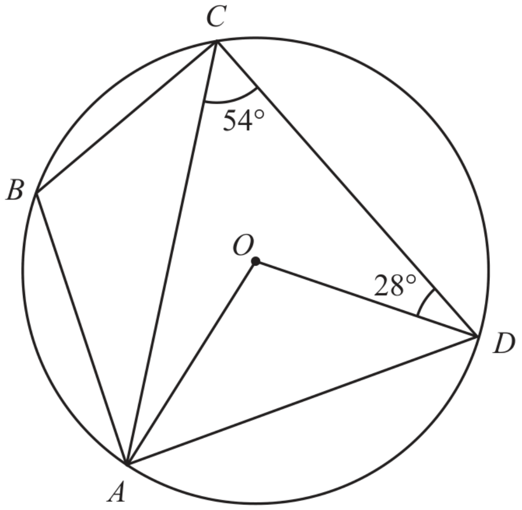

Diagram **NOT** accurately drawn

(a) (i) Work out the size of angle AOD

....
(1)

(ii) Give a reason for your answer to part (a)(i)

....

....
(1)

(b) Work out the size of angle CAO

....
(1)

(c) Work out the size of angle ABC

....
(2)

Worked solution 28

Find a central angle / Give the circle-theorem reason / Find angle CAO / Find angle ABC

MODULAR UNIT 2
5 marks

M01 – Find a central angle

Use the central-angle theorem

$$\angle AOD = 2(54°) = 108°.$$

Final answer
108 degrees

M02 – Give the circle-theorem reason

State the theorem
The angle at the centre is twice the angle at the circumference standing on the same arc.

Final answer
Angle at centre is twice angle at circumference

M03 – Find angle CAO

Use equal radii
OA=OC, so the base angles in triangle AOC are equal; with the central angle, ∠ CAO=26°.

Final answer
26 degrees

M04 – Find angle ABC

Find the adjacent angle
Use the radius/chord angles to obtain the angle at C in triangle ABC.

Use angles in a triangle

$$\angle ABC = 116°.$$

Final answer
116 degrees

Original question 29

4MA1-2H Q16

ORIGINAL QUESTION
4 marks

16 D, E, F and G are points on a circle, centre O

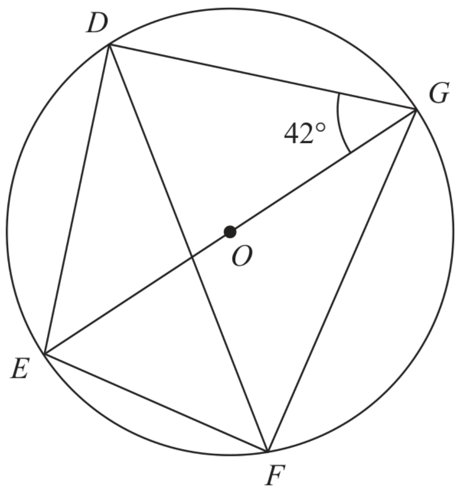

Diagram **NOT** accurately drawn

EOG is a diameter of the circle.

Angle $EGD = 42°$

Calculate the size of angle DFG
Give a reason for each stage of your working.

Angle DFG = ...°

Worked solution 29

Chapter: Circle Theorems

MODULAR UNIT 2
4 marks

Q16 – Chapter: Circle Theorems

Family: Write justified circle-angle chains • Variant: Complete a multi-theorem circle-angle chain

Method
EG is a diameter, so angle EFG=90°.
Why: An angle in a semicircle is a right angle.

Method
In triangle DEG, angle EDG=90° and angle EGD=42°, so angle DEG=48°.
Why: Angles in a triangle add to 180°.

Method
Angles DEG and DFG subtend the same chord DG, so angle DFG=48°.
Why: Angles in the same segment are equal.

Method
Therefore angle DFG is 48°.
Why: State the requested angle with reasons.

Final answer
angle DFG: 48°

Original question 31

4MA1-2H Q19

ORIGINAL QUESTION
3 marks

19

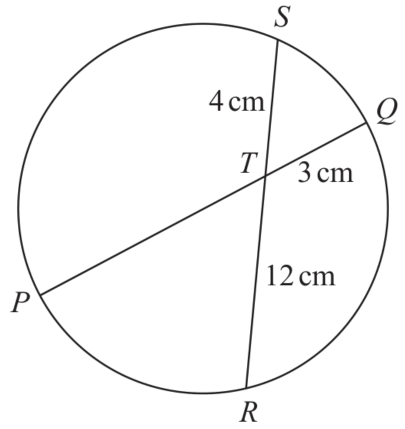

Diagram **NOT** accurately drawn

PTQ is a diameter of a circle.
RTS is a chord of the circle.

$TQ = 3\text{ cm}$ $ST = 4\text{ cm}$ $TR = 12\text{ cm}$

Calculate the radius of the circle.

.... cm

Worked solution 31

Chapter: Circle Theorems

MODULAR UNIT 2

3 marks

Q19 – Chapter: Circle Theorems

Family: Use intersecting chord and secant products • Variant: Use the internal intersecting-chords product

Method

By intersecting chords, $PT \times TQ = ST \times TR$, so $PT \times 3 = 4 \times 12$ and $PT = 16$ cm.

Why: Use the chord-intersection theorem.

Method

$PQ = PT + TQ = 16 + 3 = 19$ cm.

Why: P,T,Q are collinear and PQ is a diameter.

Method

$\text{Radius} = PQ/2 = 9.5$ cm.

Why: A radius is half a diameter.

Final answer

radius: 9.5 cm

13

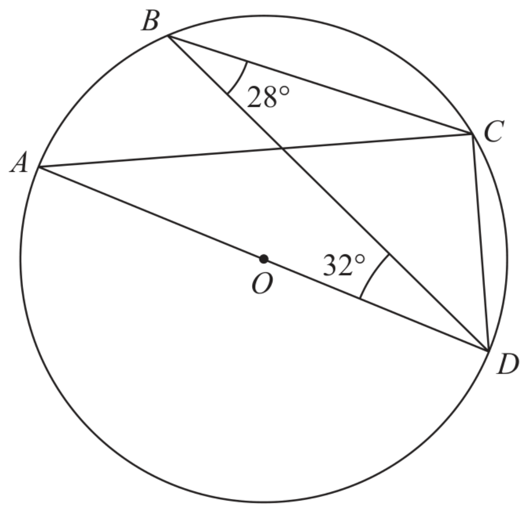

Diagram **NOT** accurately drawn

A, *B*, *C* and *D* are points on a circle, centre *O*.
AOD is a diameter of the circle.

Angle *CBD* = 28°
Angle *BDA* = 32°

Find the size of angle *BDC*.
Give a reason for each stage of your working.

....°

Worked solution

January 2020 | Question 13 | 4 marks

Family: Write justified circle-angle chains • Variant: Complete a multi-theorem circle-angle chain

Method / working

1. AD is a diameter, so angleABD=90°.
2. angleABC=90-28=62°.
3. Opposite cyclic angles give angleADC=180-62=118°.
4. angleBDC=118-32=86°.

Final answer: 86°

13 P, Q, R, S and T are points on a circle with centre O.

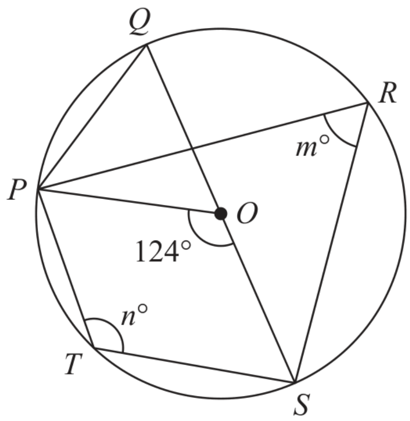

Diagram **NOT** accurately drawn

QOS is a diameter of the circle.

angle $POS = 124°$ angle $PRS = m°$ angle $PTS = n°$

(a) Find the value of

(i) m

....

(ii) n

....

(2)

(b) Find the size of angle QPO.

.... °

(1)

Worked solution

January 2021 | Question 13 | 2 marks

Family: Write justified circle-angle chains • Variant: Complete a multi-theorem circle-angle chain

Method / working

1. Angle PRS subtends the same chord PS as angle POS, giving $m=62°$.
2. Opposite angles in the cyclic quadrilateral give $n=180-62=118°$.

Final answer: $m=62°$, $n=118°$

13 P, Q, R, S and T are points on a circle with centre O.

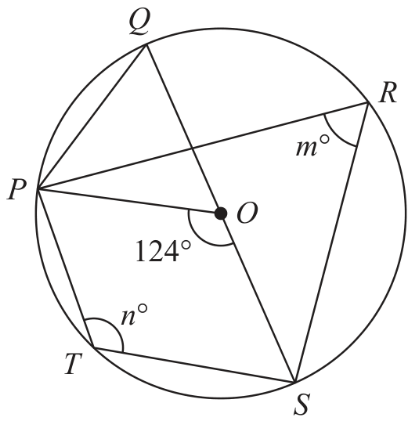

Diagram **NOT** accurately drawn

QOS is a diameter of the circle.

angle $POS = 124°$ angle $PRS = m°$ angle $PTS = n°$

(a) Find the value of

(i) m

....

(ii) n

....

(2)

(b) Find the size of angle QPO.

.... °

(1)

Worked solution

January 2021 | Question 13 | 1 marks

Family: Write justified circle-angle chains • Variant: Complete a multi-theorem circle-angle chain

Method / working

1. Triangle QOS is isosceles with QOS a diameter, so angle QPO=62°.

Final answer: 62°

15 P, Q and R are points on a circle, centre O.
TRV is the tangent to the circle at R.

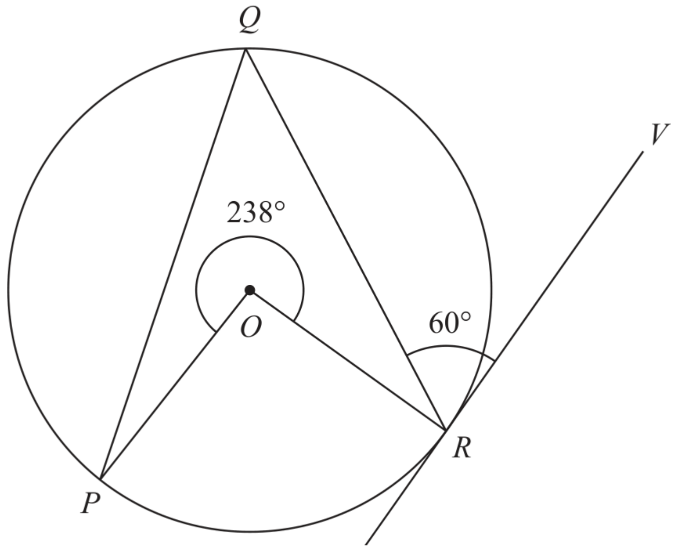

Diagram **NOT** accurately drawn

Reflex angle $POR = 238°$
Angle $QRV = 60°$

Calculate the size of angle OPQ.
Give a reason for each stage of your working.

....°

Worked solution

June 2021 | Question 15 | 4 marks

Family: Write justified circle-angle chains • Variant: Complete a multi-theorem circle-angle chain

Method / working

1. The tangent is perpendicular to the radius, giving the relevant 90° angle.
2. Use the reflex central angle 238° to obtain the minor central angle 122° and the corresponding triangle angle.
3. Use the isosceles triangle and angle sum to find the required angle.
4. Therefore angleOPQ = 31°; state the circle/tangent reasons.

Final answer: 31°

11

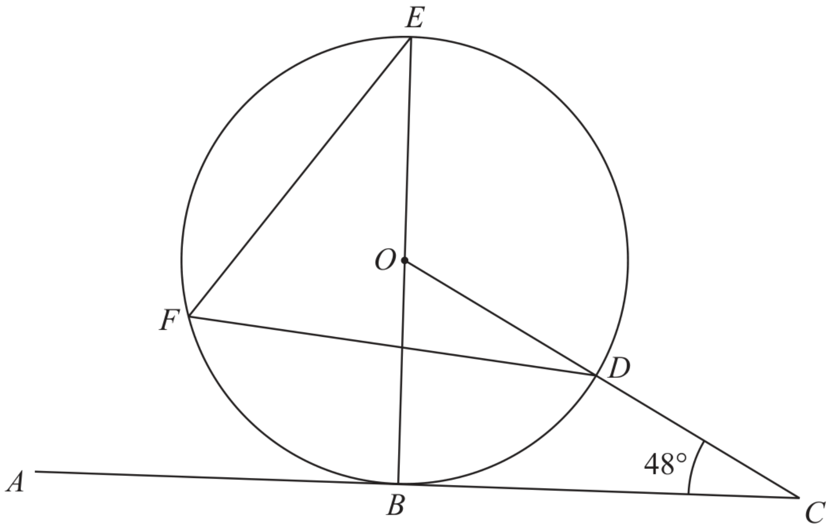

Diagram **NOT** accurately drawn

B, D, E and F are points on a circle, centre O.
ABC is a tangent to the circle.
ODC is a straight line.

BOE is a diameter of the circle.

Angle $BCD = 48°$

Find the size of angle DFE.

.... °

Worked solution

November 2021 | Question 11 | 3 marks

Family: Write justified circle-angle chains • Variant: Complete a multi-theorem circle-angle chain

Method / working

1. The tangent at B is perpendicular to the radius OB, so
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. $\angle OBC=90^\circ$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. In triangle OBC,
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. $\angle BOC=180^\circ-90^\circ-48^\circ=42^\circ$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. Since O,D,C are on a straight line,
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. $\angle BOD=42^\circ$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. BOK is a diameter, so BOE is a straight line and
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
8. $\angle DOE=180^\circ-42^\circ=138^\circ$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
9. The angle at the circumference is half the angle at the centre:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
10. $\angle DFE=\frac{138^\circ}{2}=69^\circ$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
11. Answer: 69°.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: 69°.

16

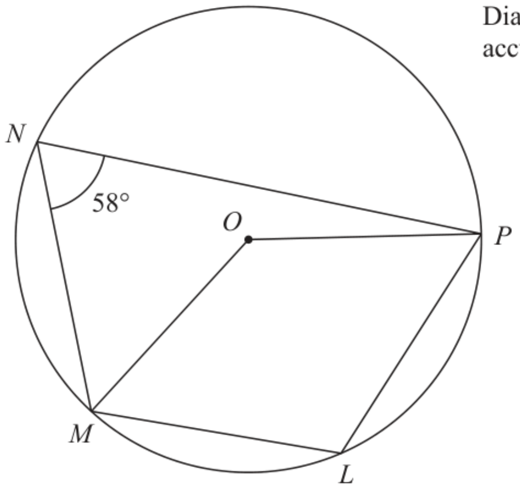

Diagram **NOT** accurately drawn

L, M, N and P are points on a circle, centre O

Angle $MNP = 58°$

(a) (i) Find the size of angle MLP

....°

(ii) Give a reason for your answer.

....

....

(2)

(b) Find the size of the reflex angle MOP

....°

(2)

(Total for Question 16 is 4 marks)

Answer reference | May-Nov 2020 | Question 16

Family: Write justified circle-angle chains • Variant: Complete a multi-theorem circle-angle chain

16	(a)(i)		122	1	B1	
	(a)(ii)		reason	1	B1	(dep on a correct answer or a correct method seen for (i)) Opposite angles in a cyclic quadrilateral sum to 180°
	(b)	$360 - 2 \times 58$ or $2 \times$ '122'		2	M1	ft from (a)
			244		A1	
						Total 4 marks

Worked solution

May-Nov 2020 | Question 16 | 4 marks

Family: Write justified circle-angle chains • Variant: Complete a multi-theorem circle-angle chain

Solution
(a) Opposite angles in cyclic quadrilateral MNPL add to 180^circ:
angle MLP=180^circ-58^circ=122^circ
(b) The angle at the centre is twice the angle at the circumference standing on the same arc MP:
angle MOP=2 x 58^circ=116^circ
The reflex angle is
360^circ-116^circ=244^circ
Answer: angle MLP=122^circ, and reflex angle MOP=244^circ.

14 A, B, C and D are points on a circle, centre O

EBF is the tangent to the circle at B

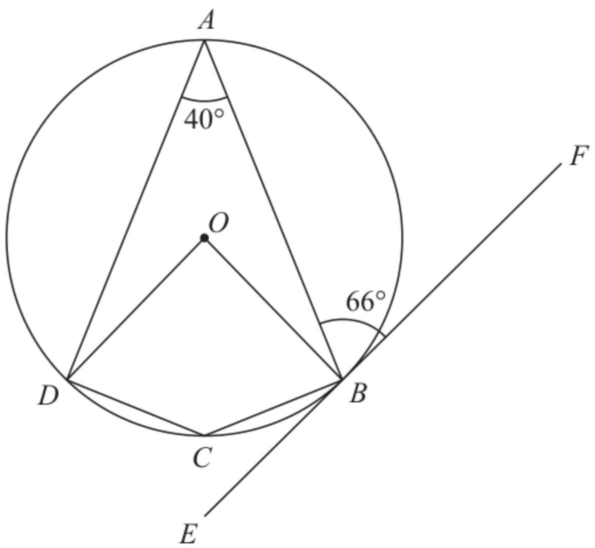

Diagram **NOT** accurately drawn

(a) (i) Work out the size of angle DCB

.... °

(1)

(ii) Give a reason for your answer to (a)(i)

....

....

(1)

(b) Work out the size of angle ADO

.... °

(3)

(Total for Question 14 is 5 marks)

Answer reference | January 2023 | Question 14

Family: Write justified circle-angle chains • Variant: Complete a multi-theorem circle-angle chain

Question		Working	Answer	Mark		Notes
14	(a)(i)		140	1	B1	
	(a)(ii)		opposite angles of a cyclic quadrilateral (add to 180°) oe	1	B1	dep on B1 in (a)(i) or seeing 180 – 40 with no contradiction oe eg angle at centre is double (2 ×) angle at circumference oe **AND** angles around a point (or point 360)
	(b)	ADB = 66 or ABO = 90 – 66 (=24) or BAO = 90 – 66 (=24) or $ODB = \frac{180-80}{2} (= 50)$ or DOB reflex = 280		3	M1	Clearly labelled in working or shown on diagram
		For 2 of: ADB = 66 or ABO = 90 – 66 (=24) or BAO = 90 – 66 (=24) or $ODB = \frac{180-80}{2} (= 50)$ DOB reflex = 280			M1	(award M2 for $360-(280+40+24)$ oe
		Correct answer scores full marks (unless from obvious incorrect working)	16		A1	
						Total 5 marks

Worked solution

January 2023 | Question 14 | 5 marks

Family: Write justified circle-angle chains • Variant: Complete a multi-theorem circle-angle chain

Official final mark-scheme pages are embedded immediately after this question.

15 A, B, C and D are points on a circle with centre O

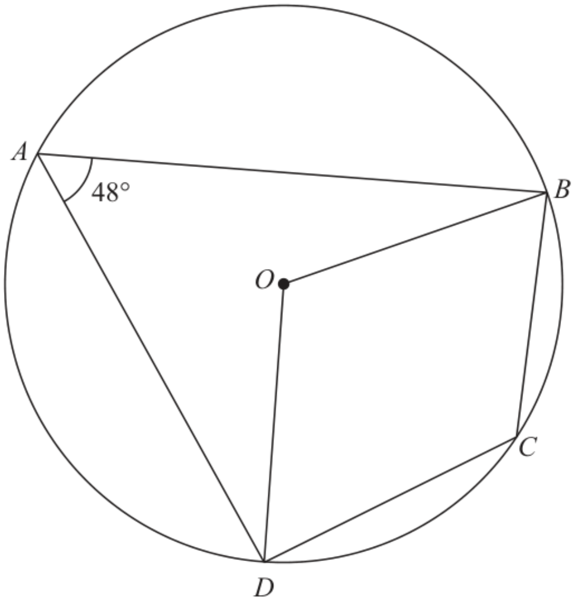

Diagram **NOT** accurately drawn

Angle $DAB = 48°$

(a) (i) Work out the size of the obtuse angle DOB

.... °

(1)

(ii) Give a reason for your answer.

....

....

(1)

(b) (i) Work out the size of angle BCD

.... °

(1)

(ii) Give a reason for your answer.

....

....

(1)

(Total for Question 15 is 4 marks)

Worked solution

November 2025 | Question 15 | 4 marks

Family: Write justified circle-angle chains • Variant: Complete a multi-theorem circle-angle chain

Solution

(a) The angle at the centre is twice the angle at the circumference standing on chord DB:

angle DOB=2 x 48^circ=96^circ

(b) Opposite angles in cyclic quadrilateral ABCD add to 180^circ:

angle BCD=180^circ-48^circ=132^circ

Answer: angle DOB=96^circ, and angle BCD=132^circ.

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION